

Emmanouil Ioannis Farsarakis

AI Performance & Systems Engineer: Physical AI, Inference Systems, HPC
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Edinburgh, UK | US, UK and EU citizen, no sponsorship required | Available from October 2026

PROFILE

AI Performance & Systems Engineer | More than a decade across computer architecture, AI infrastructure, and HPC. Most recently, in Intel's Datacenter Group, characterized LLM and vision-language-action (VLA) robotics workloads end to end and turned that analysis into architecture direction: hands-on profiling on A100 and H200, roofline-based projection across edge accelerators, and the memory-bandwidth-versus-compute tradeoffs that decide real-time serving budgets. Led the adaptation of an LLM roofline performance simulator to multimodal workloads; technical lead of a SPEC Cloud simulation effort advising AMD, Arm, and Google. Earlier, led the Harvard–Intel graph-learning acceleration research (peer-reviewed papers, US patent application, PyTorch Geometric upstream) and co-developed the low-precision machine translation that won the 2020 WNGT efficiency task (INT8 kernels later adopted into SPEC CPU). Physics and HPC background (five years at EPCC). Through-line: balancing FLOP/s against TB/s at every layer of the memory hierarchy, and being clear about what is measured versus what is projected.

PROFESSIONAL EXPERIENCE

INTEL | AI SYSTEMS & SOLUTIONS ENGINEER

(Remote) Edinburgh, UK | Nov 2023 – Sep 2026

Systems Architecture & Engineering, Datacenter Group. Characterized LLM and VLA serving ($\pi 0.x$) end to end (model structure, attention behavior, matmul sizes, BF16/INT8 execution) with hands-on profiling on A100 and H200 and roofline-based projection across edge accelerators. Led the adaptation of an LLM roofline performance simulator to multimodal workloads, including disaggregated prefill/decode serving. Hand-optimized an INT8 GEMM on a pre-production dataflow accelerator. Aggregated workload-versus-hardware data (memory capacity and bandwidth, FLOPS, FLOPS/W) across GPUs, NPUs, and dataflow accelerators that informed architecture decisions. Technical lead of a five-person SPEC Cloud simulation and performance-projection effort advising AMD, Arm, and Google.

INTEL | AI RESEARCH ENGINEER SCIENTIST

(Remote) Edinburgh, UK | Feb 2022 – Nov 2023

Coordinated research with scientists on GNNs, LLMs, and recommendation systems, connecting sparsity, memory-access patterns, and data movement to accelerator design choices. Established and led the Harvard–Intel graph-learning acceleration collaboration (about seven contributors): IEEE ISPASS 2023, TASK Quarterly 2024, US Patent Application 18/325,348, and the HGAM optimization upstreamed into PyTorch Geometric and pyg-lib.

INTEL | PLATFORM ARCHITECT

(Hybrid) Edinburgh, UK | Feb 2019 – Feb 2022

Led workload enablement and characterization for Intel PiUMA during early evaluation: simulation on internal cycle-approximate architecture simulators, profiling, microbenchmarking, and GNN scalability analysis, correlating projections against pre-production prototype hardware (IEEE ISPASS 2023). Co-developed CPU-efficient INT8 and logarithmic 4-bit Marian neural machine translation with the University of Edinburgh (K. Heafield), winning the 2020 WNGT efficiency task; the INT8 GEMM kernels were later adopted into SPEC CPU (2026).

THE UNIVERSITY OF EDINBURGH, EPCC | APPLICATIONS CONSULTANT / DEVELOPER

Edinburgh, UK | Sep 2014 – Feb 2019

HPC and data-science consulting across finance, cyber security, and exascale academia–industry collaborations. Built system-level workload characterization of ARCHER, the UK national supercomputer, and researched data-aware scheduling for Intel 3D XPoint non-volatile memory (EU NextGenIO). Modernized codes for Xeon and Xeon Phi at the Intel Parallel Computing Centre. Teaching: MSc Course Organiser, High Performance Computing Architectures (2016–2019, eight lecturers); co-developed the PRACE “Supercomputing” MOOC on FutureLearn (2017, 3,200+ registrations); led the ARCHER Software and Data Carpentry program (2015–2019, 250+ researchers trained).

SELECTED PROJECTS

Physical AI Workload Characterization (VLA and World Models)

PROFILING, ROOFLINE PROJECTION

Characterized VLA serving ($\pi 0.x$) across model structure, attention behavior, transformer-diffusion balance, and BF16/INT8 execution; hands-on profiling on A100/H200 with roofline-based projection across Jetson-class edge, NPU, and dataflow accelerators. Isolated a memory-bandwidth-bound denoising loop that inverted compute-centric optimization priorities; recommended lower-bit quantization and microscaling formats for real-time budgets.

LLM Roofline Performance Simulator (Multimodal Adaptation)

PERFORMANCE MODELING

Led the adaptation of an existing LLM roofline performance simulator to vision and VLA workloads; the simulator covers arbitrary inference configurations (architectures, parallelism, attention variants, disaggregated prefill/decode) and feeds architecture direction.

Sparse Computational Infrastructure for GNNs

RESEARCH, MANAGEMENT, ENGINEERING

Established and led a cross-org collaboration (around seven contributors across Intel and Harvard SEAS) on graph-learning acceleration for Intel® PiUMA. Owned inception, stakeholder management, software and performance optimization; resulted in ISPASS 2023, a TASK Quarterly journal paper (2024) and a US patent application.

Low-Precision CPU Inference: Marian NMT in SPEC CPU

INT8 GEMM, QUANTIZATION, CPU OPTIMIZATION

Part of the Intel–University of Edinburgh collaboration (K. Heafield) on CPU-efficient, low-precision (int8) Marian neural machine translation, winning the 2020 WNGT efficiency task. The underlying int8 GEMM kernels were adopted into the SPEC CPU benchmark in 2026, where Marian machine translation is the sole NLP workload.

SELECTED PUBLICATIONS & PATENTS

OPTIMIZING GRAPH LEARNING USING HIERARCHICAL GRAPH ADJACENCY MATRIX (HGAM)

JOURNAL | 2024

TASK Quarterly, vol. 28, no. 3

NEURAL NETWORK TRAINING AND INFERENCE WITH HIERARCHICAL ADJACENCY MATRIX

PATENT | 2023

US Patent Office - app. US18/325,348

CHARACTERIZING THE SCALABILITY OF GRAPH CONVOLUTIONAL NETWORKS ON INTEL® PIUMA

PAPER | 2023

IEEE Intl. Symposium on Performance Analysis of Systems and Software (ISPASS)

EDINBURGH’S SUBMISSIONS TO THE 2020 MACHINE TRANSLATION EFFICIENCY TASK

PROCEEDINGS | 2020

Fourth Workshop on Neural Generation and Translation

EXPERIENCES PORTING PRODUCTION CODES TO XEON PHI PROCESSORS

PROCEEDINGS | 2015

Parallel Computing: On the Road to Exascale

AWARDS AND DISTINCTIONS

Highest LINPACK

LEIPZIG, GERMANY | JUN 2014

ISC’14 STUDENT CLUSTER COMPETITION

Led the EPCC team to the “Highest LINPACK” award at ISC’14 with 3.38 TFLOP/s/kW, an estimated 4th on the June 2014 Green500 list and the first team to break 10 TFLOP/s in under 3 kW.

Highly Skilled Workforce Scholarship

EDINBURGH, UK | SEP 2013

SCOTTISH FUNDING COUNCIL

Competitive scholarship for select Edinburgh Physics and Astronomy MSc students to tackle real-world research problems.

EDUCATION

MSc in High Performance Computing

EDINBURGH, UK | AUG 2014

UNIVERSITY OF EDINBURGH

Academic Performance: Awarded with Distinction and elected Class Representative

Grade Point Average: 78%

Dissertation: “Energy Efficiency: Benefits and limitations of modern HPC architectures”

BSc Physics With Computational Science Specialization

HERAKLION, GREECE | MAR 2011

UNIVERSITY OF CRETE

Academic Performance: Top 5% of class

Grade Point Average: 77%

Teacher Assistant (TA): Introduction to C++ and Java